



Ahmedabad
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CSE623 : Machine Learning

**Machine Learning-based Estimation of Muscle
Cross-Sectional Area (ACSA) from Segmented
Ultrasound Images of Lower Limb Muscles for
Sports Performance Analysis**

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Abstract

The ultrasound image of the lower limb muscles is important in estimation of the Anatomical Cross-Sectional Area (ACSA) of muscle which is used to analyze muscle development, rehabilitation progress and sports performance. Nevertheless, the process of manual segmentation of ultrasound images is very time consuming and it is also susceptible to variation. This report investigates classical image processing and machine learning to automatically detect the two muscles (Rectus Femoris and Vastus Lateralis) in ultrasound images. Various segmentation methods like global thresholding, Otsu thresholding, adaptive segmentation, K-means clustering, graph-based segmentation, active contour models and texture-based machine learning were tested. One of these, a texture-based machine learning method based on Local Binary Pattern (LBP) features, intensity features, and a Random Forest classifier, yielded the highest performance in terms of IoU and Dice scores. The result of this segmentation will be utilized to derive geometric and statistical features of the muscle area and construct a dataset that may be utilized later in machine learning models in estimating ACSA automatically.

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1 What We Have Done

We created a pipeline in which:

- We loaded the ultrasound images of the Rectus Femoris (RF) and Vastus Lateralis (VL) muscles as well as the ground-truth segmentation masks. The images were turned to grayscale and were used as matrices in order to make image processing work easily. Given the masks, this step was done to identify the muscle region and transform it into a binary representation to make it represent the ground truth muscle region.
- Subsequently, we tested various classical segmentation methods to obtain the location of the muscle automatically. These were the global thresholding, adaptive thresholding, Otsu thresholding, K-means clustering, graph based segmentation and active contour models. Most of these techniques, however, resulted in unstable boundaries or incomplete segmentation with the presence of noise and low contrast in ultrasound images.
- To improve the quality of segmentation, we adopted a machine learning method that uses texture. Under this approach, we obtained several pixel-based characteristics like normalized intensity, Gaussian-blurred intensity, Local Binary Pattern (LBP) texture characteristics, gradient magnitude, and the spatial coordinates. We then trained a Random Forest classifier using these features to categorize every pixel as either a muscle or background pixel. We then used morphological operations to clean the mask following prediction. Intersection over Union (IoU) and Dice score were used to assess the segmentation performance and this measure resulted in the best performance of all the methods that were tested.

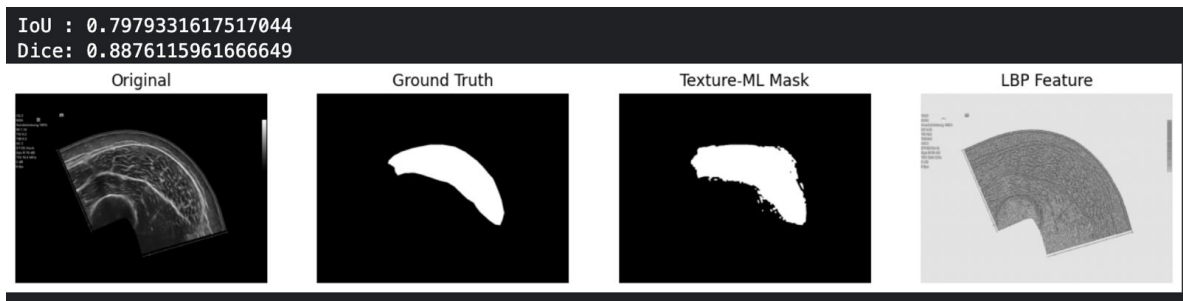


Figure 1: Segmentation results with texture based Random Forest Classifier

2 What We Aspire to Do

- Our next steps will be to extract meaningful features of the segmented muscle region. We plan on using an element-wise multiplication of the segmentation mask and the image matrix to extract the region of interest (ROI) in the ultrasound image based on the predicted mask. Based on this ROI, the geometric features, area, perimeter, compactness and axis based measures will be calculated to characterize the shape of the muscle.
- Besides the use of geometric features, the statistical and texture-based features of the ultrasound intensity values within the muscle region will be computed. They can be mean intensity, variance and other texture measures that describe internal structure of the muscle tissue. All the features that were extracted will be then arranged in structured feature vectors by sample of ultrasound.
- Lastly, these feature vectors shall be assembled together into a dataset that shall be analysed further under machine learning. The data will be divided into a training and a test set and the regression models will be investigated to determine the Automatic estimation of the Anatomical Cross-Sectional Area (ACSA). Appropriate error metrics will be used to assess model performance and correlation between extracted features and ACSA will be computed as well.

References

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